

Semantic-Aware Joint Source-Channel Coding for Mobile Edge Networks

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Question

How much task accuracy can be preserved when a mobile-edge system sends semantic features rather than raw features?

64-dimensional features are compressed to 8 dimensions and transmitted over Rayleigh fading.

Seeded result

At 2 dB: semantic accuracy 90.83%, raw baseline 100%, dimension ratio 12.5%.

This is a PCA baseline, not a final Deep JSCC claim. Public image/task data and learned training are next.